Class 10 NCERT Maths

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Chapter 5: Arithmetic Progressions

CHAPTER 5: ARITHMETIC PROGRESSIONS

A simplified, detailed guide in Hinglish
For students who missed Class 7-8-9
Based on the official NCERT English Medium textbook

SECTION 1: FOUNDATION

1.1 SEQUENCE KYA HOTI HAI?

SEQUENCE = numbers ki ek list jo kisi specific pattern follow karti hai.

EXAMPLES:

1, 3, 5, 7, 9, ...   (odd numbers)
2, 4, 8, 16, 32, ...   (powers of 2)
1, 4, 9, 16, 25, ...   (squares)
5, 10, 15, 20, ...   (multiples of 5)

Har number ko TERM kehte hain - 1st term, 2nd term, etc.

1.2 ARITHMETIC PROGRESSION (AP) - DEFINITION

AP = wo sequence jisme har term, PREVIOUS term mein ek FIXED number ADD karke milti hai.

Wo fixed number ko COMMON DIFFERENCE (d) kehte hain.

EXAMPLES OF AP:

2, 5, 8, 11, 14, ...   (d = 3)
10, 7, 4, 1, -2, ...   (d = -3)
5, 5, 5, 5, ...   (d = 0)
-1, 2, 5, 8, ...   (d = 3)

NOT AP:

1, 4, 9, 16, ...   (squares - diff. badalti hai)
2, 4, 8, 16, ...   (multiplication, not addition)

1.3 AP KE NOTATIONS

Notation	Matlab
a (or a1)	First term
d	Common difference
a_n	nth term
n	Number of terms
S_n	Sum of first n terms
l	Last term

1.4 COMMON DIFFERENCE D KAISE NIKAALEIN?

FORMULA: $d = a_n - a_{(n-1)}$
(any term minus previous term)

EXAMPLE: AP: 5, 9, 13, 17, ...

d = 9 - 5 = 4
d = 13 - 9 = 4 (ok)
d = 17 - 13 = 4 (ok)

IMPORTANT: Saare consecutive differences same hone chahiye, warna AP nahi hai.

1.5 GENERAL FORM OF AP

----

a, a+d, a+2d, a+3d, a+4d, ...

So:
1st term = a
2nd term = a + d
3rd term = a + 2d
4th term = a + 3d
nth term = a + (n-1)d

SECTION 2: nth TERM FORMULA

2.1 THE FORMULA (IMPORTANT)

$$a_n = a + (n-1)\,d$$

Yahan:
a_n = nth term (jo nikaalna hai)
a = first term
d = common difference

n = term number

2.2 EXAMPLES

EXAMPLE 1: AP 2, 5, 8, 11, ... ka 20th term

a=2, d=3, n=20
a_20 = 2 + (20-1)(3)
= 2 + 57
= 59

Answer: 59

EXAMPLE 2: AP ka 4th term 0 hai aur 11th term -7
hai. AP nikaalo.

a + 3d = 0 ...(1)
a + 10d = -7 ...(2)

(2) - (1): 7d = -7 -> d = -1
From (1): a = -3d = 3 -> a = 3

So AP: 3, 2, 1, 0, -1, -2, -3, -4, ...

EXAMPLE 3: 100, 95, 90, ... ka kaunsa term -50 hai?

a=100, d=-5, a_n = -50

-50 = 100 + (n-1)(-5)
-150 = -5(n-1)
n - 1 = 30
n = 31

So 31st term -50 hai.

SECTION 3: SUM OF FIRST n TERMS

3.1 THE FORMULAS (DONO YAAD RAKH)

Formula 1 (a aur d known ho):

S_n = (n/2) x [2a + (n-1) d]

Formula 2 (a aur last term l known ho):

$S_n = (n/2) \times (a + l)$

jahan l = last term = a_n.

YAAD RAKH: $a_n = a + (n-1)d$, isliye dono formulas connected hain.

3.2 EXAMPLES

EXAMPLE 4: 2, 5, 8, 11, ... ke first 20 terms ka sum

$a=2, d=3, n=20$

$$S_{20} = (20/2) \times [2(2) + 19(3)]$$
$$= 10 \times [4 + 57]$$
$$= 10 \times 61$$
$$= 610$$

EXAMPLE 5: 1+2+3+...+100 (Gauss formula)

AP: 1, 2, 3, ..., 100

$a=1, l=100, n=100$

$$S_{100} = (100/2) \times (1 + 100)$$
$$= 50 \times 101$$
$$= 5050$$

EXAMPLE 6: AP 2, 5, 8, ... ka kitne terms ka sum
155 hoga?

$a=2, d=3, S_n = 155$

$$155 = (n/2)[4 + 3(n-1)]$$
$$310 = n[3n + 1]$$
$$3n^2 + n - 310 = 0$$

Splitting ($a \times c = -930$, sum = 1):
31 x -30 -> sum = 1, product = -930 (ok)
$(3n + 31)(n - 10) = 0$

$n = 10$ (positive)

So 10 terms.

SECTION 4: WORD PROBLEMS

EXAMPLE 7: Theatre seats

"First row mein 20 seats. Har next row mein 2
seats zyada. 30th row mein kitne seats?"

$a=20$, $d=2$, $n=30$
$a_{30} = 20 + 29 \times 2 = 78$

Answer: 78 seats.

EXAMPLE 8: Salary problem

"Starting salary Rs.8000. Har saal Rs.500 increment.
10 saal ki total earnings?"

AP: 8000, 8500, 9000, ... ($a=8000$, $d=500$)

$S_{10} = (10/2) \times [2(8000) + 9(500)]$
$= 5 \times [16000 + 4500]$
$= 5 \times 20500$
$=$ Rs. 1,02,500

Q AND A TIME

Q1. Inn sequences mein se kaunsi AP hai? d bata.
(a) 2, 4, 6, 8, ...
(b) 1, 4, 9, 16, ...
(c) -3, -1, 1, 3, ...
(d) 1, 3, 9, 27, ...

Q2. AP: 7, 13, 19, 25, ... ka 50th term nikaal.

Q3. AP ka 10th term 32 hai aur 20th term 72 hai.
AP nikaal.

Q4. AP: 1, 4, 7, 10, ... ke first 25 terms ka sum.

Q5. AP 9, 17, 25, ... mein kaunsa term 105 hoga?

Q6. "Ek admi 8 din mein har din 5 km zyada distance
cover karta hai. Pehle din 3 km. 8 dino mein
total km?"

SUMMARY

1. AP: Sequence with constant common difference d.

2. General form: a, a+d, a+2d, a+3d, ...

3. nth term: $a_n = a + (n-1) d$

4. Sum formulas:
   $S_n = (n/2)[2a + (n-1)d]$ (d known)
   $S_n = (n/2)(a + l)$ (l known)

5. d nikaalna: any term minus previous term.

6. Word problems: identify a, d, n; pick formula.
